Unit 2 Review · Roll for It

Code name _____ Period ____

Unit 2 Cells · LT 2.1–2.5 · Mr. Schwebach rolls the die; everyone works the roll that comes up.

How it runs. One roll, about four minutes: one minute to talk with a partner, three to write. Then check against the board and fix in a second color. **A 6 is two parts**, because transport is about a third of the test. A repeat roll means the 6 again, or the roll you trust least.

Cell theory and how science works

LT 2.1 · 5 of 32 points

a. Ana writes: “If bread is kept in a warm cabinet, mold will grow on it faster than on bread in the fridge.” Circle one: **hypothesis** · **theory** · **law**. What tells you?

b. Mark each statement **T** if it is part of cell theory, **F** if not. Then circle the letter.

___ I. Every cell has a nucleus.
___ II. All living things are made of one or more cells.
___ III. New cells come only from cells that already exist.
___ IV. Every cell is about the same size.

A. I and II B. II and III C. III only D. II, III and IV

c. Under the microscope, a yeast cell grows a bump that pinches off into a new yeast cell. Which part of cell theory does that support?

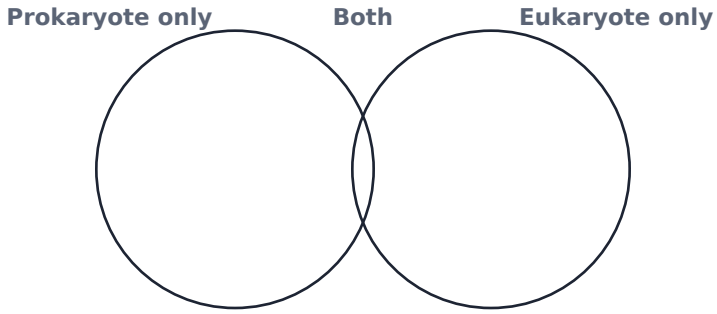
d. Cell theory has been revised since the 1800s. What made the revisions possible?

Prokaryote, eukaryote, or both

LT 2.2 · part of 7 points

a. Write each word in the right part of the diagram. Every region is a point, and **Both** counts.

ribosomes · nucleus · DNA · cell membrane · mitochondria · bacteria · Golgi apparatus · no membrane-bound organelles



b. Circle the **two** structures a plant cell has that an animal cell does not: mitochondria · cell wall · ribosome · chloroplast · nucleus

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Organelles working together

LT 2.2 · part of 7 points

a. Number the steps 1–5 for a protein the cell ships out.

___ the Golgi apparatus packages and labels it
___ a ribosome on the rough ER builds it
___ a vesicle carries it to the cell membrane
___ it moves through the rough ER
___ the vesicle fuses with the membrane and releases it

b. Choose **TWO** correct pairs. Mark every choice T or F first, then circle two letters.

___ A. nucleus: holds DNA and directs the cell ___ B. ribosome: stores water ___ C. mitochondrion: releases energy from food ___ D. cell wall: builds proteins ___ E. vacuole: copies DNA

c. A heart muscle cell works without stopping. Which organelle should it have the most of, and why?

Levels of organization and homeostasis

LT 2.3 · 5 of 32 points

a. Order these from smallest to largest: organ · organism · cell · organ system · tissue

_____ → _____ → _____ → _____ → _____

b. A paramecium swims, eats and reproduces all by itself. A cell from your stomach lining cannot survive on its own. Which one is unicellular? What is the stomach cell part of, and why can it specialize?

c. A bean plant's water loss in one hour, at four air temperatures:

Air temperature (°C)	15	25	30	40
Water lost (mL)	2.0	5.5	7.1	2.4

Where did water loss peak? _____ What did the stomata most likely do at 40 °C, and why does that count as homeostasis?

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The cell membrane

LT 2.4 · 5 of 32 points

a. Sketch a short stretch of membrane, two layers thick. Label: **water-loving head · water-fearing tail · channel protein · cholesterol · carbohydrate chain**. Write water on both sides.

b. In the fluid mosaic model, *fluid* means _____ and *mosaic* means _____.

c. Oxygen crosses the membrane easily; most sugar cannot cross without help. One phrase for that: s_____ p_____.

Second chance

Pick the roll you trust least. Write its number here: _____. Redo one part of it from memory, without looking back, then check it.

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Transport: two parts, both every time

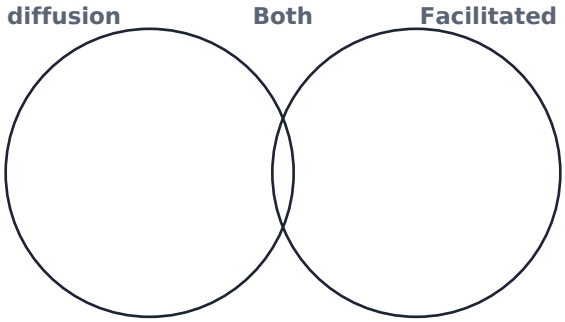
LT 2.5 · 10 of 32 points

6A. Sort each process: diffusion · osmosis · facilitated diffusion · a protein pump moving ions from low to high · endocytosis · exocytosis

Needs energy (ATP):

No energy needed:

Now the diagram. Write each phrase in its region: **uses a channel or carrier protein · moves from high to low concentration · oxygen slipping between phospholipids · uses no ATP**



6B. Red blood cells go into three cups. Name each solution (**hypotonic · isotonic · hypertonic**), then draw the water arrows and the cell after ten minutes.



Cup 1: pure water _____

Cup 2: 0.9% salt (same as the cell) _____

Cup 3: 10% salt _____

A U-tube is split by a membrane that lets water through but not sugar. The left arm holds 3% sugar, the right arm 9% sugar. Water moves from the _____ side to the _____ side. The _____ side rises. The process is _____.

Five moves for the test

1. **Choose TWO?** Mark every choice T or F before you circle. Both have to be right for the point.
2. **Diagrams and sorts.** Every box is its own point, and "both" is a box. Fill every one.
3. **I, II, III items.** Judge each numeral alone, write T or F beside it, then find the letter that matches.
4. **Tables and graphs.** Say the trend in your own words (rises, peaks, drops) before you read the choices.
5. **Osmosis.** Water moves toward the side with more dissolved stuff. Find the higher sugar or salt first.

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